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8.1 Trauma Assessment and Transport Considerations

The assessment and management of the injured patient must be efficient and organized. The assessment should ensure critical findings are found and reliably addressed in a prioritized manner. The approach to trauma care may be organized into a three-step process comprised of an initial impression, primary assessment, and secondary head to toe assessment.

NOTE: **The ABCDE approach is used to organize the trauma assessment, not mandate the steps be done rigidly. Medical Crews must use their judgment and prioritize treatment based on the clinical circumstances.**

8.1.1 Initial Impression

As part of the initial impression, systematically consider how the mechanism of injury and operational situation might affect the patients condition and clinical course during transport. Once at the bedside, take in an overview of the patients appearance and the treatment taking place. Make a determination about the patients general appearance, treatment in process, and immediate needs.

8.1.2 Primary Assessment

Perform a primary assessment to identify life threats related to airway, oxygenation, ventilation, circulation, and neurological injury.

NOTICE **Obvious life threatening external hemorrhage should be immediately controlled as it is found using the techniques described in** hemorrhage control**.**

Airway

- If the patient has an advanced airway, confirm its placement, position, and function.
- Evaluate responsiveness and the ability to protect airway
- In the responsive patient, assess airway patency by asking the patient to talk and validate the patient's ability to maintain their own airway.
- Assess for injuries that may lead to airway obstruction including unstable facial fractures, expanding neck hematoma, or bleeding in the airway, and facial burns.
- Observe for airway obstructions such as the tongue, blood, vomit, or edema.
- If the patient is unable to manage their own airway, the Medical Crew should establish an airway with manual maneuvers, suction and basic adjuncts. Advanced airway management should follow resuscitation of the patient.

Breathing

- Expose and examine the chest and neck for injures and abnormalities
- Note the work of breathing and ventilation rate
- If the respiratory effort is ineffective or the rate is < 10 or > 30 per minute, begin bag mask ventilation
- Auscultate lung sounds for presence and equality

Oxygen—Empiric Administration for Trauma Scene

Adult

- FiO2 100% or 15-25 L/min

Pediatric

- Refer to adult dosing

⚠ CAUTION **Trauma patients may have blood loss and reduced oxygen carrying capacity. Acute and critically ill multi-trauma patients should get high flow oxygen and down titration of FiO2 should be avoided.**

Open pneumothorax

Open chest wounds should be immediately sealed with one of the following techniques to restore respiratory mechanics.

- Commercial vented chest seal (Asherman chest seal preferred)
- Three or 4 sided occlusive dressing. Any air tight material and tape will work. Taping on three sides may allow for a vent.

NOTE: **The palm of a gloved hand applying pressure right on top of the wound may be used as a temporary measure while a more permanent seal is prepared.**

⚠ CAUTION **Once the wound is sealed, monitor for the development of a tension pneumothorax and be prepared to vent the chest by removing the occlusive dressing or performing a thoracostomy.**

Tension pneumothorax

Tension pneumothorax is more common in ventilated than awake patients and patients with chest trauma. Difficult ventilation and hypoxia followed by hypotension in the newly intubated and ventilated trauma patient is axiomatic for tension pneumothorax. Slower development is possible, but less likely in patients receiving positive pressure ventilation.

Signs and Symptoms—Ventilated Patients

NOTE: **In order of likelihood**

- Rapid onset
- Immediate and progressive decrease in Sp02
- Immediate reduction in cardiac output and hypotension
- High ventilation pressures
- Chest hyper-expansion
- Decreased lung sounds
- Subcutaneous emphysema
- Venous distension

NOTICE **A low threshold for performing needle or simple thoracostomy should be maintained in ventilated trauma patients.**

Signs and Symptoms—Awake Patients

Awake patients are less likely to present with a tension pneumothorax and the progression may be insidious.

NOTE: **In order of likelihood**

- Chest pain
- Tachycardia
- Decreased lung sounds
- Hypoxia

Cross reference		
Clinical Procedures	17	3-15 - Needle Thoracostomy
Clinical Procedures	17	3-8 - Simple and Tube Thoracostomy

Circulation

- Assess skin color, vital signs and peripheral pulses
- Establish vascular access and resuscitate as per the Shock and CV instability Guideline

Hemorrhage Control

Systematically apply bleeding control techniques to all likely sources of hemorrhage

Direct Pressure

- Application of direct pressure to the wound with gauze is the standard procedure to control bleeding

Wound Packing

- Uncontrolled bleeding in deep wounds of the extremities or junction between extremities, may be packed.

NOTE: **Tightly packing the wound and applying as much pressure as possible to the bleeding vessel is the key to hemorrhage control.**

Pressure Bandage

- A pressure dressing may be applied to maintain bleeding control after direct pressure or wound packing
- The SWAT T may be used as pressure dressing

Tourniquet

- A Combat Application Tourniquet (CAT) or SOF Tactical Tourniquet (SOF TT) should be used immediately for life threatening extremity wounds, or if direct pressure or wound packing is ineffective or impractical.

Splinting

- Align and expediently splint long bone fractures

Pelvic Binder

- Apply a pelvic binder if a pelvic fracture is likely or suspected

Bandage and dressing in place with ongoing uncontrolled bleeding

- Remove the dressing down to the injury site
- Remove clots and definitively identify the bleeding site
- Control bleeding as indicated in this guideline

Dressing

- Apply a dry sterile dressing to the wound after bleeding is controlled
- Splint or immobilize as needed to reduce the risk of dislodging the packing, dressing, or tourniquet
- Reassess all dressings and wounds periodically for re-bleeding

NOTE: **If available, hemostatic gauze should be used for direct pressure, pressure dressings, and wound packing**

Cross reference		
Clinical Care Guidelines	23	4-20 - Shock & Cardiovascular Instability
CAT Combat Application Tourniquet Gen 7 DFU	0	[to document]

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Disability

- If the patient has an altered level of consciousness, check blood glucose levels and treat hypoglycemia immediately.
- Prior to administering neuromuscular blocking agents (NMB), obtain a baseline level of consciousness, Glasgow Coma Scale (GCS) score, and motor sensory function

Spine Assessment

Anticipate cervical spine injury and apply spinal motion restriction (SMR) in patients with significant blunt traumatic injuries who meet the criteria for SMR. The medical crew should evaluate the need for or adequacy of any SMR already in place.

- Conduct a thorough physical assessment of the spine and back, focusing on palpating accessible areas to the greatest degree possible.
- Elicit a report of the pre-transport assessment (PTA) findings from the sending providers prior to SMR application.

Exposure

- Remove clothing as necessary to perform assessment
- Monitor and control temperature

Brain Injuries

- Where practicable, keep the head in a midline position and elevate the head of the bed (HOB) by 15–30 degrees.
- If the patient is intubated, maintain End-Tidal CO2 (EtCO2) levels between 35-45 mmHg.
- If there is clinical evidence of herniation syndrome (such as deteriorating level of consciousness, unilateral dilated pupil, widening pulse pressure, varying respiratory patterns, or bradycardia), end-tidal CO2 (EtCO2) should be maintained at 30–35 mmHg.

⚠ CAUTION

The odds of death are up to six times greater for out-of-hospital patients with traumatic brain injury who experience hypotension or hypoxia. High FiO2 and aggressive resuscitation may be necessary to treat and prevent hypoxia and hypotension.

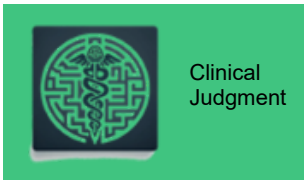
Traumatic Brain Injury

Positioning

- Whenever possible, keep the patient's head in a midline position and elevate the head of the bed (HOB) between 15 and 30 degrees.

Monitoring

- **Glasgow Coma Scale (GCS):** Obtain a baseline GCS and continuously monitor throughout transport.
- **Pupillary Response:** Record baseline pupillary size and reactivity, and continue to monitor throughout transport.
- **Respiratory Rate and Pattern:** Monitor the respiratory rate and pattern consistently to ensure stability.
- **Oxygenation:** Maintain oxygen saturation (SpO2) levels above 90% to ensure adequate tissue oxygenation.
- **Ventilation:** For intubated patients, the target end-tidal CO2 (EtCO2) should be maintained between 35-45 mmHg.
- **Blood Pressure:** Maintain systolic blood pressure (SBP) within the target range of 100-110 mmHg.



In trauma cases, the goal for systolic blood pressure (SBP) is typically set at 90 mmHg. However, for traumatic brain injury (TBI) patients with altered mental status, who are at high risk for increased intracranial pressure (ICP), the SBP target is adjusted to 100-110 mmHg to ensure adequate cerebral perfusion pressure. This contrasts with non-traumatic intracerebral hemorrhage, where the underlying mechanisms differ significantly and the SBP goals are substantially higher. For specific hemorrhagic stroke guidelines, see 7-7 - Hemorrhagic Stroke.

Patients with traumatic brain injury (TBI) who experience hypotension or hypoxia outside of hospital settings have a mortality risk up to six times higher than those without these complications. To effectively manage and prevent hypoxia and hypotension, use high fractional inspired oxygen (FiO2) levels to ensure adequate oxygenation and employ aggressive resuscitation techniques to stabilize blood pressure.

Herniation Syndrome Monitoring

Monitor for signs of herniation syndrome.

Brain Herniation Syndrome

Brain herniation may occur due to expanding mass lesions associated with conditions such as intracranial hemorrhage, stroke, abscess, tumors, and hydrocephalus.

- If practicable, medical crews should manage herniation syndrome collaboratively with treating providers or PHI medical directors to ensure decisions are informed by comprehensive medical insights and tailored to the patient's specific needs.
- In cases where medical direction is unavailable, medical crews should follow these guidelines to treat herniation syndrome.

Signs of Brain Herniation Syndrome

- **Neurogenic Breathing Patterns:** Monitor for patterns including Cheyne-Stokes, hyperventilation, and ataxic breathing.
- **Abnormal Pupillary Responses:** Look for abnormalities such as asymmetrical dilation and fixed pupils.
- **Abnormal Motor Responses:** Assess for signs such as decerebrate posturing (extensor posturing), decorticate posturing (flexor posturing), hemiparesis, hemiplegia, and flaccidity.
- **Cushing Reflex:** This reflex is characterized by bradycardia, respiratory depression, and hypertension.

Neurocritical Care

Herniation Syndrome Treatment

If herniation syndrome is suspected:

- Adjust the ventilator settings to achieve an end-tidal CO2 (EtCO2) level between 30-35 mmHg.

  NOTE: **Hyperventilation therapy is recognized as the fastest method to reduce intracranial pressure (ICP).**

- Administer osmotic therapy.

*This is a RESCUE therapy for herniation syndrome used to acutely lower ICP.

Sodium Chloride 3%

Adult

- 250 mL IV over 5-15 min

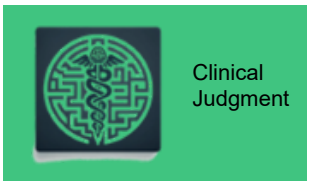
Ped

- 3–5 mL/kg IBW IV over 5-15 min

NOTE: **This is a RESCUE therapy for herniation syndrome, used to acutely lower intracranial pressure (ICP)**

Analgesia and Sedation

- The sedation goal is to achieve a Richmond Agitation-Sedation Scale (RASS) score of -2.
- Ketamine is the preferred agent for scene responses and for managing undifferentiated trauma.



For interfacility transports, propofol may be considered for patients without contraindications, such as hemodynamic instability. Propofol is beneficial as it reduces the cerebral metabolic rate of oxygen consumption (CMRO2), diminishes cerebral blood flow and volume, and consequently lowers intracranial pressure (ICP). Additionally, its anticonvulsant properties make it a suitable option for managing neurological conditions during patient transfers.

Seizure Prophylaxis

Seizure Prophylaxis

- TXA may be given for TBI if requested by regional trauma services.

  NOTE: **TXA for TBI studies show limited benefit but no harm.**

8.1.3 Secondary Assessment and Transport Considerations

Abdominal

- Suspected/known diaphragmatic rupture: signs and symptoms may include absent or diminished breath sounds on the affected side, bowel sounds may be heard in the chest cavity when intestinal contents have herniated into the pleural space, respiratory distress/hypoxia.
- Eviscerations should be covered with saline-soaked gauze pads. Do not attempt to push the organs back into the abdomen.
- If a diaphragmatic rupture is suspected, consider placement of a gastric tube to reduce the possibility of aspiration and gastric dilatation, especially for patients with herniation.

Pregnancy

Assessment of patient to include fetal status. Suspect occult intra-abdominal injury.

- Perform fetal assessment
- Transport patient in left lateral decubitus position – late pregnancy > 20 weeks

Orthopedic and Soft Tissue

- Frequently evaluate distal circulation, motion, sensation (CMS) of affected extremities
- Distal CMS must be evaluated before and after every manipulation and after splinting
- SpO2 monitoring may be used to evaluate distal perfusion on the injured limb
- Deformed long bone fractures should be returned to normal alignment
- Splints should stabilize the joint above and below the fracture site

NOTE: **Injured extremities may be expediently splinted by whole body stabilization in the anatomic position on a back board, sled, or cot**

Amputation

- Splint if practicable
- Care of amputated part:
  - Rinse amputated part
  - Cover/wrap amputated part with sterile NS dampened gauze or towel
  - Place amputated part into a dry, sealed plastic bag
  - Place sealed bag into a container filled with ice and water

Avulsion

- Rinse gross debris from wound and put the avulsed tissue in its normal position over wound site
- Dress the wound

Impaled Object

- **Do not** remove object
- Stabilize object in place with bulky dressing

Pelvic Fracture or Clinically Unstable Pelvis

- Stabilize the pelvis with a SAM pelvic sling, other commercial device or sheet wrap.
- Perform a neurovascular exam before and after any extremity manipulation or splinting.

CAUTION **Do not place a urinary catheter in any patient with a suspected pelvic fracture or bleeding from the urinary meatus or perianal area. Doing so can cause further injury, bleeding or urethral damage, leading to serious complications.**

NOTICE **Perform a neurovascular exam before and after any extremity manipulation or splinting.**

NOTICE **Notify the receiving facility of any extremity injury associated with absent distal pulses.**

Femur Fracture

- Apply steady traction and rotation to return to normal alignment.
- The affected leg may be expediently splinted using the other leg or a back board.

Femur Traction Device

A traction device/splint may be applied to well-aligned midshaft femur fractures

Indications	Contraindications	PHI Traction Devices
• Mid shaft femur fracture	• Hip fracture/injury • Knee dislocation/injury • Ankle injury	Kendrick Traction Device (KTD) Slishman Traction Splint (STS) Optimum Traction Device (OTD)

General Fracture Care

Fractures with Severe Angulation

- Apply steady traction and rotation to return the extremity to normal alignment

Presumed Fracture with Absent Distal Pulse

- Apply steady traction and rotation to return to normal alignment
- Attempt realignment up to a maximum of 2 attempts

Dislocation or Joint Injury with Absent Distal Pulse

- Apply steady traction and rotation to return to normal alignment
- Attempt realignment up to a maximum of 2 attempts

CAUTION **Traction, rotation or realignment should NOT TO BE performed on elbow injuries.**

Open Fractures

Any skin opening near a suspected fracture should be considered an open fracture.

- Control bleeding, realign, and splint as indicated
- Cover wound with saline soaked gauze

Antibiotics for open fractures

Early administration of antibiotics greatly reduces morbidity associated with open fractures

One of the following antibiotics may be given

NOTE: **Select only one of the following antibiotics**

Cefazolin—Open Fracture

Adult

- ≥ 50 kg: 2 gm IV over 3–5 min, IM
- < 50 kg: 1 gm IV over 3–5 min, IM

Pediatric

- 25 mg/kg IV over 3–5 min, IM

Clindamycin

Adult

- 900 mg IV over 30 min

Ped

- 10 mg/kg IV over 10-30 min
- Max dose 900 mg

NOTE: **Infusion rate should not to exceed 30 mg/min**

NOTE: **Use with patients allergic to cephalosporins or penicillin**

Ocular Injuries

- Remove contacts PRN
- Apply a loose sterile dressing to the affected eye
- Cover with eye shield PRN
- Cover both eyes PRN to prevent eye movement in affected eye

8.1.4 Crush Injury

Severe crush injury results from physical trauma to the torso or extremities due to sustained compression by an external crushing force. Once the compressive force is released, the systemic effects of crush syndrome may occur, resulting in organ dysfunction or death.

- Crush syndrome can develop after 1 hour of entrapment.
- Effective medical care is required to reduce the risk of kidney damage, cardiac arrhythmia, and death.

Assessment

Indications for crush syndrome in a pre-hospital environment include:

- Torso or mid or proximal extremities entrapped and crushed for greater than 1 hour

Treatment

IV Fluids

The administration of intravenous crystalloids is the primary method to prevent hyperkalemia, arrhythmias, and ventricular fibrillation caused by the release of potassium from injured tissues in the pre-hospital environment.

Sodium Chloride 0.9%

Adult

Initial Bolus

- 40 mL/kg IBW
- Maximum 2 L

After Initial Bolus

- 20 mL/kg IBW over 1 hour
- Maximum 1 L

Maintenance

- 1.5 times maintenance infusion rate

  NOTE: **May substitute with LR**

Ped

- Refer to adult dosing

Calcium

- Initially treat ECG changes consistent with hyperkalemia including peaked T waves, widening QRS or arrhythmias with calcium
- Perform a 12 lead ECG as soon as it is practicable
- Continue hyperkalemia treatement as per Hyperkalemia Protocol

Calcium Gluconate

Adult

- 3 gm IV over 2–10 minutes
- Repeat every 10 minutes PRN up to 3 times

Pediatric

- 60 mg/kg IV over 2–10 minutes
- Repeat in 10 minutes PRN
- Maximum single dose 3 gm

NOTE: **Faster rate of IV administration may be indicated in cardiac arrest, cardiovascular instability and crush injures.**

NOTE: **Flush the IV line after administration**

Calcium Chloride

Adult

- 1 gm IV over 2–10 min
- Repeat every 10 minutes PRN up to 3 times

Pediatric

- 20 mg/kg IV over 2–10 min
- Repeat in 10 minutes PRN
- Maximum single dose 1 gram

NOTE: **Faster rate of IV administration may be indicated in cardiac arrest, cardiovascular instability and crush injuries**

NOTE: **Flush the IV line after administration**

Special Considerations

- If hypotensive and suspect hemorrhagic etiology, initiate blood infusion per PHI protocol

Tourniquet Use

- Tourniquet use to involved extremity only if unable to fluid resuscitate on scene

8.1.5 Thermal Injuries

The initial priority when caring for a burn patient is to ensure responder safety and stop the burning process. Otherwise, the primary ABCDE assessment of the burn trauma patient is unchanged from any other trauma patient. During the secondary assessment, several key burn assessment and treatment actions must be performed.

Airway and Breathing

Early protection of the airway may be necessary. Examine the oropharynx for signs of inflammation, mucosal injury, soot, and edema. The larynx is vulnerable to obstruction from exposure to heat. Indications for early intubation due to thermal injury include:

- Hoarseness, stridor, accessory respiratory muscle use, sternal retraction
- Large total body surface area burn > 40%–50%
- Extensive and facial burns
- Burns inside the mouth
- Difficulty swallowing

NOTE: **Evidence of inhalation injury may be subtle and not appear until several hours after injury.**

Resuscitation

Burn resuscitation is required to replace the ongoing losses from capillary leak due to inflammation. Taking care not to over-resuscitate, initiate or continue burn resuscitation fluids for deep partial and full-thickness burns larger than 20% TBSA.

NOTE: **Dextrose containing** pediatric maintenance fluids **should be given in addition to the burn resuscitation fluids.**

Fixed IV Rates

Aged based fixed IV fluid rates may be used during pre-hospital and early transport phases of care.

- 5 years old and younger: 125 ml LR per hour
- 6-13 years old: 250 ml LR per hour
- 14 years and older: 500 ml LR per hour

Body Weight/Surface Area Burned IV Rates

When practicable, initiate, transition to, or continue burn resuscitation.based on the burns surface area and depth.

- Adult—2 mL of LR x TBW x % TBSA
- Pediatric—3 mL of LR x TBW x % TBSA
- One-half of the volume is given over the first 8 hours
- The remaining one-half is administered during the succeeding 16 hours

Resuscitation Physiologic Targets

Titrate fluid resuscitation to urine output if possible

- Adults: 30–50 mL/hr
- Pediatric 1 mL/kg/hr

If a urinary catheter is not in place, physiologic endpoints may be used to assess resuscitation effectiveness and guide therapy

- HR
- SBP
- Cap Refill
- Mentation

Carbon Monoxide Poisoning

CO poisoning should be suspected in thermal injury patients with a history of inhalation injury or direct measurement of an abnormal carboxyhemoglobin (HbCO). Signs and symptoms of include:

- Burns to face or neck
- Carbonaceous sputum, singed nasal hairs, soot in oral cavity
- History of burn injury occurring in a confined space such as a structure or vehicle fire
- Carboxyhemoglobin level greater than 10%

Monitoring

- Zoll X series monitors using rainbow sensors may be used to measure SpO2, SpCO, and SpMet monitor treatment.
- Conventional two-wavelength pulse oximeters are not accurate when COHgb is present.

NOTE: **Zoll LNCS single use and reusable sensors are capable of measuring SpO2 only.**

Carbon monoxide poisoning should be presumed, and the highest FiO2 administered in any patient with suspected inhalation injury. High concentration oxygen should be continued and down titration of FiO2 based on SpO2 is prohibited until a normal serum blood carboxyhemoglobin level is confirmed ruling CO poisoning.

Oxygen

Adult

- FiO2 100% or 15–25 L/min

Pediatric

- Refer to adult dosing

Cyanide Poisoning

Cyanide poisoning should be suspected in fire victims with inhalation injury. If available, hydroxocobalamin antidotal therapy may be given.

Hydroxocobalamin

Adult

- 5 gm IV infusion over 15 min
- May repeat 1 time
- Max total dose 10 gm

Pediatric

- 70 mg/kg IV infusion over 15 min
- Max single dose 5 gm
- Max total dose 10 gm

NOTE: **May give undiluted IV push in cardiac arrest**

NOTE: **History of smoke inhalation, removal from fire, Ingestion, inhalation, or dermal exposure to cyanide combined with neurological, cardiac or respiratory complaint.**

Hydroxocobalamin—Advisories

- Administration of the second dose may range from 15 minutes to 2 hours based on patient condition
- The recommended diluent is 0.9% Sodium Chloride
- Use the diluent included in the kit or follow kit instructions if a diluent is not included
- Use a separate intravenous line for administration due to multiple incompatibilities

8.1.6 Electrical Burns

TBSA >20%

- LR 4 mL/kg/TBSA for any electrical injury
- Obtain EKG
- Insert urinary drainage catheter if practicable
- Assess for risk or presence of compartment syndrome
- Carefully assess for contact point wounds

NOTE: **Cutaneous burns do not reflect of the extent of deep tissue injury in an electrical burn**

Urine Output Goal Adjustments

TBSA ≤ 20%

- Adult—MIVF rate to produce urine output of 30–50 mL/hr
- Pediatric—MIVF rate to produce urine output of 1 mL/kg/hr

Myoglobinuria

If myoglobinuria is present

- Adult—150–200 mL/hr to produce urine output of 75–100 mL/hr
- Pediatric—1.5–2 time MIVF rate to produce urine output of 1.5–2 mL/kg/hr

Pediatric MIVF

Use the 4-2-1 rule to calculate pediatric maintenance IV fluid flow rates

- 2-10 kg @ 4ml/kg/hr
- 11-20 kg @ 40ml/hr + 2ml/kg for each kg > 10 kg
- >20kg @ 60ml/hr + 1ml/kg for each kg >20 kg
- Maximum flow rate 100 mL / hr

Pediatric MIVF Maintenance IV Fluid Flow Rate					
Handtevy Color	Purple	White	Blue	Orange	Green
Age (yr)	1	3	5	7	9
Weight (kg)	10	15	20	25	30
Maintenance (ml/hr)	**40**	**50**	**60**	**65**	**70**

NOTICE **Pediatric MIVF IV maintenance fluids should be administered via an infusion pump**

Chemical Burns

- Protect yourself with appropriate PPE
- Remove patient clothing, jewelry, and contact lenses that could potentially contain the agent
- Decontaminate the patient by irrigating of the affected areas for 20 or more minutes using a shower or hose.
- Do not soak.
- This includes the eyes if exposed.
- This is the only intervention that will stop the burning process
- Carefully brush away dry chemicals prior to irrigation

NOTE: **Do not transport a person with chemical burns that has not been thoroughly decontaminated**

Burn Transport Considerations

- Determine the time of injury
- Monitor distal neurovascular function with circumferential burns to extremities
- Remove all clothing and expose the patient for assessment
- Remove all jewelry as quickly as possible
- Pain control may be difficult and require higher and more frequent doses of analgesics
- Monitor urine output closely and place a urinary catheter if possible
- Cover head and maintain a warm cabin temperature in flight
- Cover burns with clean dry dressings
- Keep the patient warm

⚠ CAUTION **If moist dressings were applied prior to arrival, replace them with clean dry dressings or sheets prior to transport. Wet dressings contribute to hypothermia.**

8.2 Hypothermia

8.2.1 Treatment

- Measure core temperature continuously
- Decrease heat loss by removing wet clothing and maintaining a warm environment
- Cover with warm dry blankets
- Warm all IV fluids
- Handle patient gently

Mild Hypothermia

32-35 C

- Fully awake, shivering and very uncomfortable, tachycardia, mild amnesia
- Warm fluids as tolerated

Moderate Hypothermia

28-32 C

- Altered level of consciousness, minimal shivering, confusion, and heart rate decreasing
- Avoid stimulation, thermal blanket, and and warm oxygen.

Severe Hypothermia

Less than 28 C

- Obtunded, unconscious, no shivering, bradypnea, severe bradycardia.
- Active re-warming technics (heated oxygen, warm gastric lavage, ECMO)

8.2.2 Frostbite

- Consider hypothermia and manage appropriately.
- Decrease heat loss by moving patient to a heated environment, handle patient gently, and cover with warm blankets.
- Do not rub frostbitten areas, bandage with loose sterile dressings.
- Protect the affected areas from trauma or partial thawing (If unable to totally thaw and keep thawed, withhold the thawing process until arrival at receiving facility).

8.2.3 Heat-related Illness

- Move patient to a cool environment and attempt to cool. Remove as many clothes as possible and apply cold pack, cool wash cloths to lateral chest walls, groin and axillary areas.

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